

EX NO:1 Butterworth IIR Filters

1.2. Design and Analysis of a Butterworth IIR High-Pass Filter Using
MATLAB.

Objective:

1. To determine the minimum filter order and cutoff frequency, required to satisfy the design constraints.
2. To design and analyze the magnitude response, phase response, characteristics of the Butterworth IIR High-pass filter for the given passband and stopband specifications.
3. To analyze the impulse response, step response and group delay characteristics of the Butterworth IIR High-pass filter for the given passband and stopband specifications.
4. To examine the pole-zero plot and verify the stability and performance of the designed High-pass filter.

Procedure:

1. Open MATLAB and create a new script file.
2. Enter the filter specifications: passband ripple (R_p), stopband attenuation (R_s), passband frequency (F_p), stopband frequency (F_s), and sampling frequency (F_{sampling}).
3. Normalize the passband and stopband frequencies with respect to the Nyquist frequency ($F_{\text{sampling}}/2$).
4. Use the `buttord()` function to determine the minimum filter order and cutoff frequency for the Butterworth high-pass filter.
5. Design the IIR Butterworth high-pass filter using the `butter()` function and obtain the numerator and denominator coefficients.
6. Display the filter order and cutoff frequency.
7. Compute the frequency response of the filter using the `freqz()` function.
8. Plot the magnitude response and verify that low-frequency components are attenuated while high-frequency components are passed.
9. Plot the phase response and analyze the phase characteristics.
10. Generate the pole-zero plot using the `zplane()` function and verify that all poles lie inside the unit circle.
11. Create a test signal consisting of low-frequency and high-frequency sinusoidal

components.

12. Apply the designed high-pass filter to the test signal using the filter() function.

13. Plot and compare the input and filtered output signals.

14. Analyze the filter performance in terms of frequency selectivity and stability.

15. Verify that the designed filter satisfies the given specifications.

Program (Matlab Code):

```
clc; clear
```

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DATE:

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```
Fs=10000; Fp=3000; Fst=2000;
```

```
[N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);
```

```
[b,a]=butter(N,Wn,'high');
```

```
% Magnitude & Phase
```

```
freqz(b,a)
```

```
% Impulse Response
```

```
figure
```

```
[h,n]=impz(b,a,50);
```

```
stem(n,h), grid on
```

```
title('Impulse Response')
```

```
% Step Response
```

```
figure
```

```
stepz(b,a)
```

```
title('Step Response')
```

```
% Group Delay
```

figure

grpdelay(b,a)

title('Group Delay')

% Pole-Zero Plot

figure

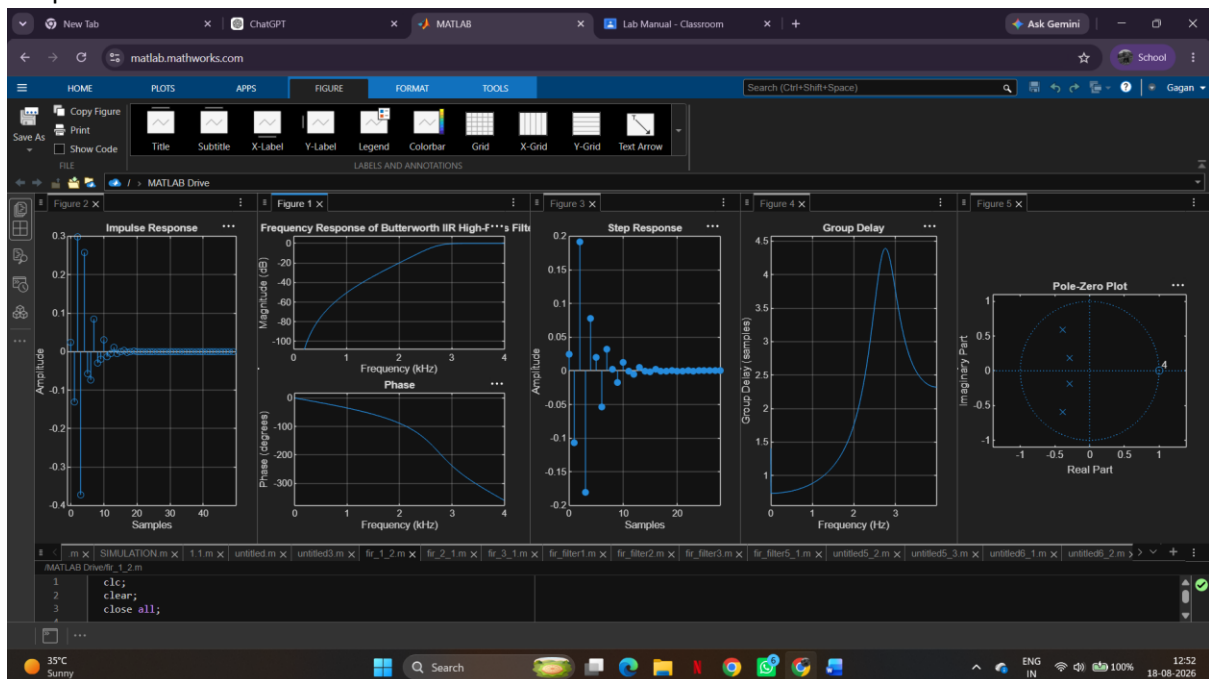
zplane(b,a)

title('Pole-Zero Plot')

fprintf('Minimum Filter Order (N) = %d\n',N);

fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);

Output:



Minimum Filter Order (N) = 12

Cutoff Frequency (Wn) = 0.5807

Result:

- 1.The minimum filter order and cutoff frequency satisfying the given specifications were determined successfully.
- 2.The magnitude and phase responses of the Butterworth IIR High-pass filter were obtained and analyzed.
- 3.The impulse response, step response, and group delay characteristics were evaluated successfully.

4. The pole-zero plot confirmed the stability and proper performance of the designed filter.